Assignment 11

Chapter 9

4. Prove Lemma 9.3.
5. Consider $f(\mathbf{Q}, \mathbf{t})$ in the BA algorithm for computing the rate-distortion function.
   (a) Show that for fixed s and $\mathbf{t}$, $f(\mathbf{Q}, \mathbf{t})$ is minimized by
   $$Q(\hat{x}|x) = \frac{t(\hat{x})e^{sd(x,\hat{x})}}{\sum_{\hat{x}'} t(\hat{x}')e^{sd(x,\hat{x}')}}.$$
   (b) Show that $f(\mathbf{Q}, \mathbf{t})$ is convex.

Chapter 10

1. Prove Propositions 10.3 and 10.8.
2. Show that the joint pdf of a multivariate Gaussian distribution integrates to 1.
3. Show that $|K| > 0$ in (10.18), the formula for the joint pdf of a multivariate Gaussian distribution.
4. Show that a symmetric positive definite matrix is a covariance matrix.
5. Let
   $$K = \begin{bmatrix} 7/4 & \sqrt{2}/4 & -3/4 \\ \sqrt{2}/4 & 5/2 & -\sqrt{2}/4 \\ -3/4 & -\sqrt{2}/4 & 7/4 \end{bmatrix}.$$
   (a) Find the eigenvalues and eigenvectors of K.
   (b) Show that K is positive definite.
   (c) Suppose K is the covariance matrix of a random vector $\mathbf{X} = [X_1\ X_2\ X_3]^\top$.
      i. Find the coefficient of correlation between X_i and X_j for $1 \leq i < j \leq 3$.
      ii. Find an uncorrelated random vector $\mathbf{Y} = [Y_1\ Y_2\ Y_3]$ such that $\mathbf{X}$ is a linear transformation of $\mathbf{Y}$.
      iii. Determine the covariance matrix of $\mathbf{Y}$.